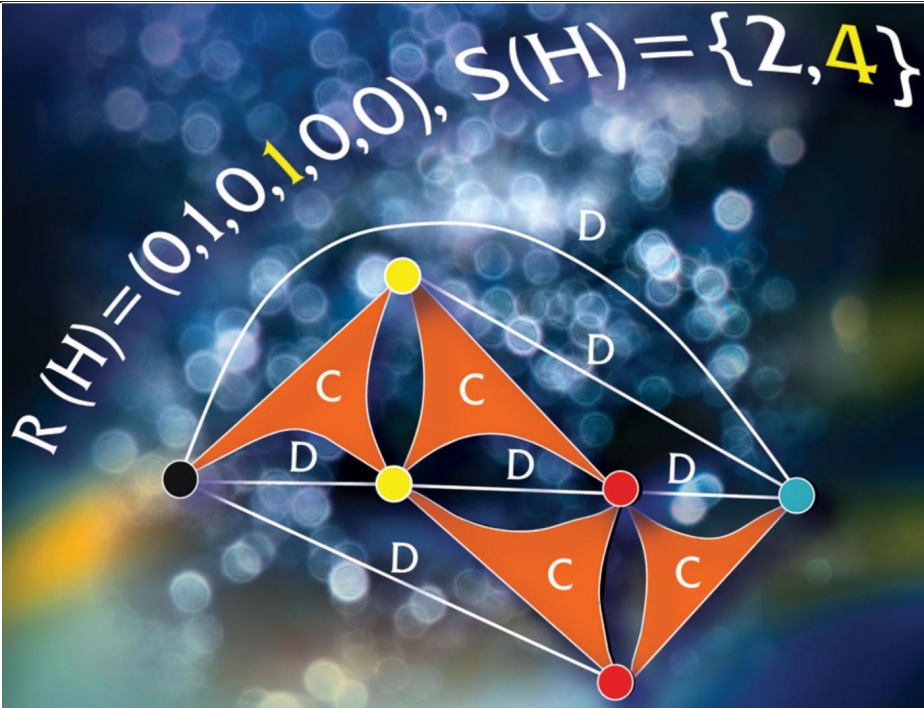
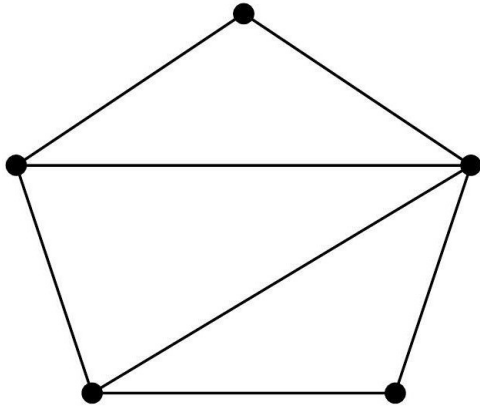
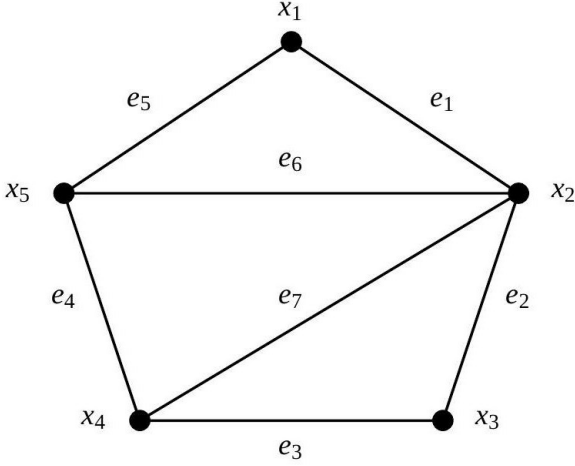
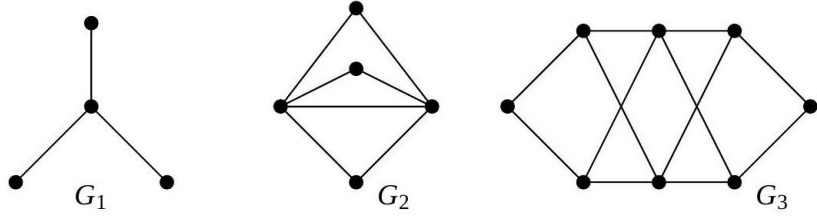
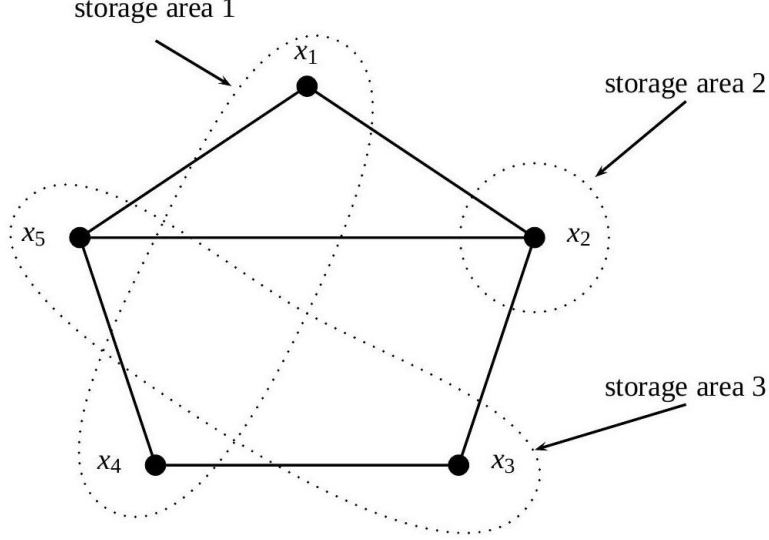
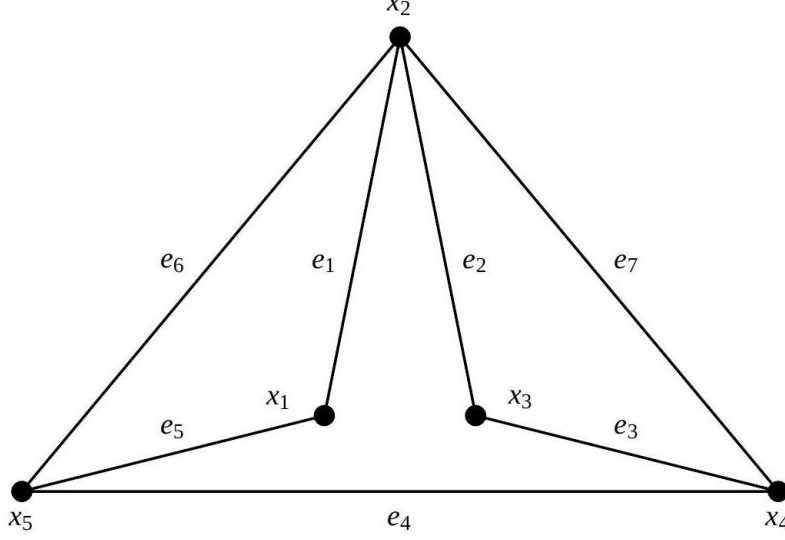
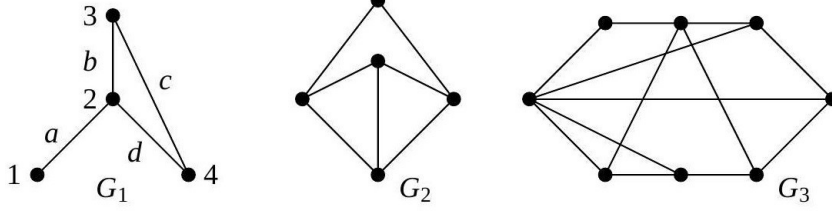
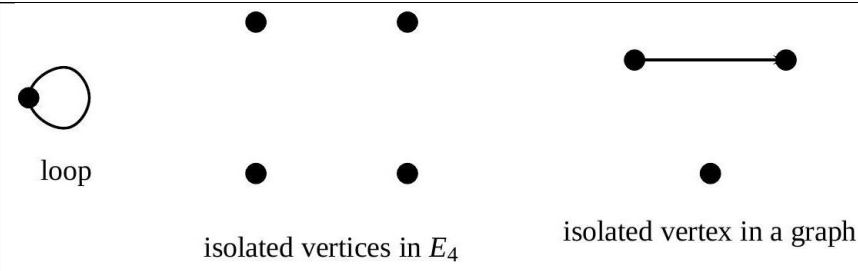
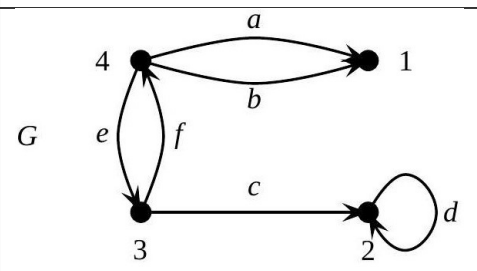
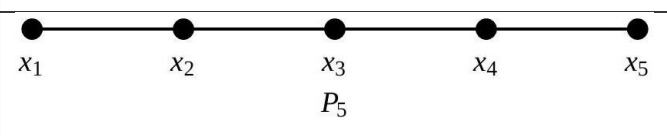
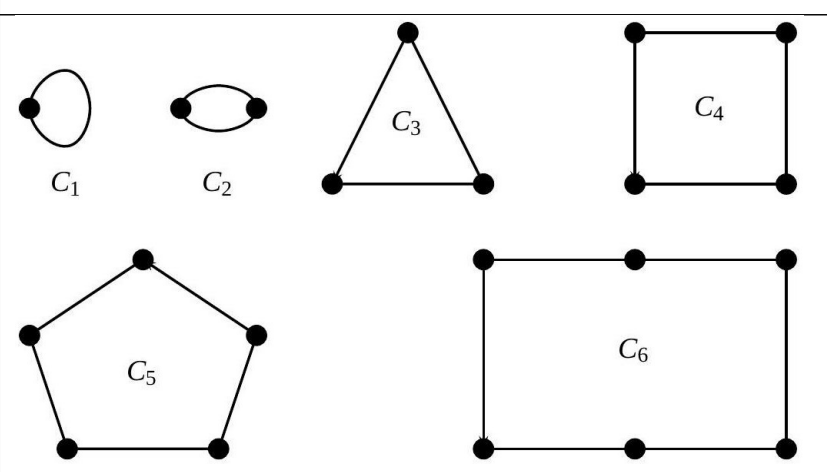
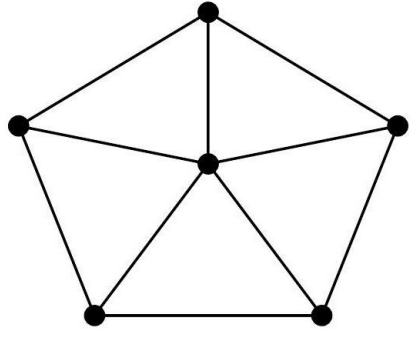
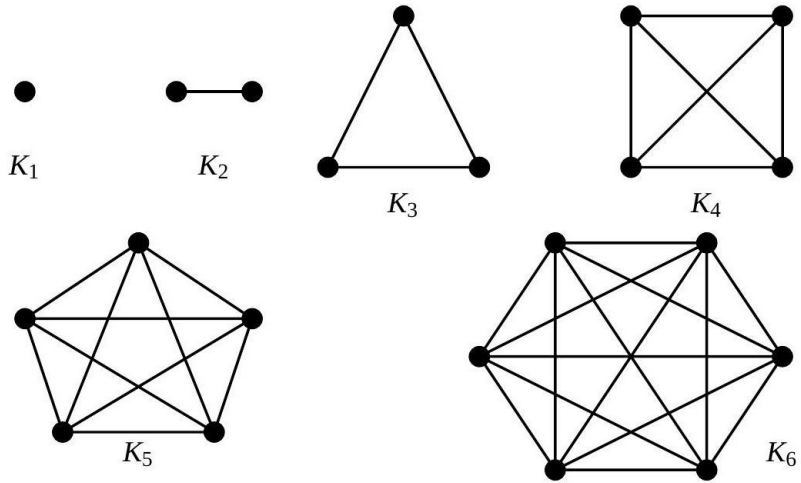
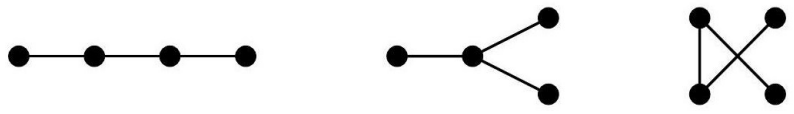
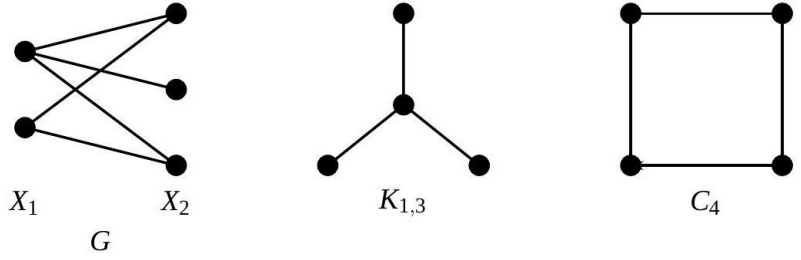
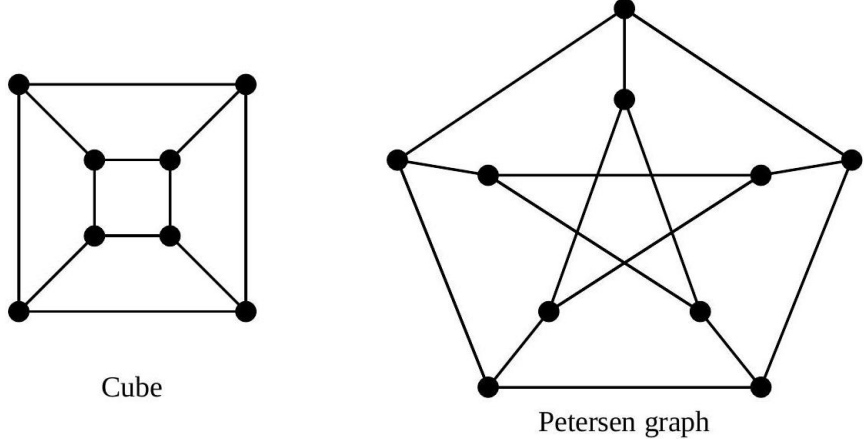


Image regions

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voloshin-hypergraph_ DIA_0003	22	—	—	—	

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voloshin-hypergraph_ DIA_0005	25	—	—	—	
voloshin-hypergraph_ DIA_0006	29	—	—	—	
voloshin-hypergraph_ DIA_0007	31	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0008	32	—	—	—	 loop isolated vertices in E_4 isolated vertex in a graph multiple edges arcs
voloshin-hypergraph_ DIA_0009	32	—	—	—	
voloshin-hypergraph_ DIA_0010	33	—	—	—	 G
voloshin-hypergraph_ DIA_0011	34	—	—	—	 P_5
voloshin-hypergraph_ DIA_0012	34	—	—	—	 G_1 G_2
voloshin-hypergraph_ DIA_0013	35	—	—	—	 C_1 C_2 C_3 C_4 C_5 C_6

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0016	36	—	—	—	
voloshin-hypergraph_ DIA_0017	37	—	—	—	
voloshin-hypergraph_ DIA_0018	38	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0020	39	—	—	—	
voloshin-hypergraph_ DIA_0021	40	—	—	—	
voloshin-hypergraph_ DIA_0022	41	—	—	—	
voloshin-hypergraph_ DIA_0023	42	—	—	—	

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voloshin-hypergraph_ DIA_0026	43	—	—	—	
voloshin-hypergraph_ DIA_0027	44	—	—	—	
voloshin-hypergraph_ DIA_0028	44	—	—	—	

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voloshin-hypergraph_ DIA_0031	48	—	—	—	
voloshin-hypergraph_ DIA_0032	49	—	—	—	
voloshin-hypergraph_ DIA_0033	51	—	—	—	

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voloshin-hypergraph_ DIA_0035	52	—	—	—	
voloshin-hypergraph_ DIA_0036	52	—	—	—	
voloshin-hypergraph_ DIA_0037	53	—	—	—	
voloshin-hypergraph_ DIA_0038	56	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0040	59	—	—	—	
voloshin-hypergraph_ DIA_0041	60	—	—	—	
voloshin-hypergraph_ DIA_0042	61	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0043	64	—	—	—	The diagram illustrates a relationship between two sets, X and Y. Set X is on the left and contains two subsets: S and $X - S_0$. Set Y is on the right and contains two subsets: $N_G(S)$ and $N_G(S_0)$. Two edges, G_A and G_B, are shown. Edge G_A connects the subset S in X to the subset $N_G(S_0)$ in Y. Edge G_B connects the subset $X - S_0$ in X to the subset $N_G(S)$ in Y.
voloshin-hypergraph_ DIA_0044	65	—	—	—	The diagram shows two sets, X and Y. Set X contains subsets S and A. Set Y contains subsets $N_{G_1}(S)$ and B. A horizontal dashed line divides the diagram into two regions: G_1 above and G_2 below. Two edges are shown: one connecting S to $N_{G_1}(S)$ and another connecting A to B.
voloshin-hypergraph_ DIA_0045	66	—	—	—	The diagram depicts a grid of points connected by lines. There are four rows and six columns of points. Horizontal lines connect all points in each row. Vertical lines connect the first, third, fifth, and sixth columns of points. This structure forms a series of nested rectangles and a central open space.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0046	68	—	—	—	
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voloshin-hypergraph_ DIA_0048	70	—	—	—	
voloshin-hypergraph_ DIA_0049	72	—	—	—	
voloshin-hypergraph_ DIA_0050	74	—	—	—	
voloshin-hypergraph_ DIA_0051	75	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0054	76	—	—	—	
voloshin-hypergraph_ DIA_0055	77	—	—	—	
voloshin-hypergraph_ DIA_0056	78	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0059	81	—	—	—	
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0061	84	—	—	—	
voloshin-hypergraph_ DIA_0062	84	—	—	—	
voloshin-hypergraph_ DIA_0063	86	—	—	—	
voloshin-hypergraph_ DIA_0064	87	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0065	88	—	—	—	Diagram illustrating the construction of a hypergraph from a K_5 graph. The K_5 graph is shown with vertices 1, 2, 3, 4, 5. Two paths are shown: "4 inside" and "4 outside". The "4 inside" path leads to a graph with vertices 1, 2, 3, 4, 5 and a face for 5. The "4 outside" path leads to a graph with vertices 1, 2, 3, 4, 5 and a face for 5.
voloshin-hypergraph_ DIA_0066	91	—	—	—	Diagram illustrating the construction of a hypergraph from a $K_{3,3}$ bipartite graph. The $K_{3,3}$ graph is shown with vertices 1, 2, 3, 4, 5, 6. Two paths are shown: "3 inside" and "3 outside". The "3 inside" path leads to a graph with vertices 1, 2, 3, 4, 5 and a face for 6. The "3 outside" path leads to a graph with vertices 1, 2, 3, 4, 5 and a face for 6.
voloshin-hypergraph_ DIA_0067	92	—	—	—	Diagram illustrating the construction of a hypergraph from a graph G_1. The graph G_1 is shown with vertices 1, 2, 3, 4, 5 and a face for 5. The graph G_2 is shown with vertices 1, 2, 3, 4, 5 and a face for 5.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0068	93	—	—	—	
voloshin-hypergraph_ DIA_0069	97	—	—	—	
voloshin-hypergraph_ DIA_0070	99	—	—	—	
voloshin-hypergraph_ DIA_0071	100	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0072	101	—	—	—	Diagram illustrating the decomposition of a hypergraph G into two sub-hypergraphs G_1 and G_2. G consists of two vertices x and y connected to a set of internal nodes. G_1 is identical to G but with a dashed edge between x and y. G_2 consists of a single vertex xy connected to the same set of internal nodes. The equation $G = G_1 + G_2$ is shown.
voloshin-hypergraph_ DIA_0073	102	—	—	—	Diagram illustrating the decomposition of a tree G into two sub-hypergraphs G_1 and G_2. G consists of three vertices x, y, and z connected in a tree structure. $G_1 = K_3$ is a triangle with vertices x, y, and z. $G_2 = K_2$ is a path with vertices y and xz. The equation $G = G_1 + G_2$ is shown.
voloshin-hypergraph_ DIA_0074	103	—	—	—	Diagram illustrating a hierarchical tree structure G. The root node G branches into G_1 and G_2. G_1 branches into G_3 and G_4, and G_2 branches into G_5 and G_6. Each of these nodes further branches into two leaf nodes. Below the tree, a sequence of nodes $K_{i_1}, K_{i_2}, \dots, K_{i_s}$ is shown, connected by dots.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0075	104	—	—	—	
voloshin-hypergraph_ DIA_0076	105	—	—	—	
voloshin-hypergraph_ DIA_0077	107	—	—	—	$E_3 - 2E_2 + E_1 \Rightarrow P(G, \lambda) = \lambda^3 - 2\lambda^2 + \lambda = \lambda(\lambda - 1)^2$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0078	108	—	—	—	
voloshin-hypergraph_ DIA_0079	109	—	—	—	
voloshin-hypergraph_ DIA_0080	111	—	—	—	
voloshin-hypergraph_ DIA_0081	112	—	—	—	
voloshin-hypergraph_ DIA_0082	113	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0083	115	—	—	—	
voloshin-hypergraph_ DIA_0084	117	—	—	—	
voloshin-hypergraph_ DIA_0085	119	—	—	—	
voloshin-hypergraph_ DIA_0086	120	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0087	121	—	—	—	
voloshin-hypergraph_ DIA_0088	122	—	—	—	
voloshin-hypergraph_ DIA_0089	124	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0090	125	—	—	—	The diagram shows a transformation from a single edge (two vertices connected by a line) to a more complex graph structure. The transformation is indicated by an arrow $\Rightarrow$. The resulting graph has five vertices and several edges, with one cycle highlighted by a dotted line.
voloshin-hypergraph_ DIA_0091	125	—	—	—	The diagram shows a complex graph structure with 10 vertices and many edges. The graph is highly connected, with many cycles and a central vertex connected to several others.
voloshin-hypergraph_ DIA_0092	126	—	—	—	The diagram shows two graph structures, G and $\bar{G}$, with vertices labeled 1, 2, 3. G is a graph with 5 vertices and 6 edges, with vertices 1, 2, 3 labeled. $\bar{G}$ is the complement of G, with 5 vertices and 4 edges, with vertices 1, 2, 3 labeled. The graphs are shown with dotted lines around them.
voloshin-hypergraph_ DIA_0093	129	—	—	—	The diagram shows two graph transformations. The top transformation shows a graph G with 5 vertices and 6 edges, with vertices labeled a, b, c, d, e, transforming into a graph $L(G)$ with 6 vertices and 10 edges, with vertices labeled a, b, c, d, e. The bottom transformation shows a graph with 5 vertices and 6 edges, with vertices labeled 1, 2, 3, 4, 5, transforming into a graph with 6 vertices and 10 edges, with vertices labeled 1, 2, 3, 4, 5, 6. The bottom transformation is labeled "edge coloring" and "vertex coloring".

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0094	130	—	—	—	
voloshin-hypergraph_ DIA_0095	131	—	—	—	
voloshin-hypergraph_ DIA_0096	133	—	—	—	
voloshin-hypergraph_ DIA_0097	134	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0098	136	—	—	—	G
voloshin-hypergraph_ DIA_0099	140	—	—	—	
voloshin-hypergraph_ DIA_0100	141	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0101	145	—	—	—	The image displays three network flow diagrams arranged vertically, connected by downward arrows, illustrating the progression of a flow augmentation algorithm. Each diagram features a source node u and a sink node v, with intermediate nodes x_1, x_2, x_3, x_4. Initial flow F_0: - Edges and flows: $u \rightarrow x_1$ (0,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (1,7), $x_1 \rightarrow x_3$ (1,1), $x_2 \rightarrow v$ (1,6), $x_3 \rightarrow v$ (0,2), $x_4 \rightarrow x_3$ (1,4). Flow F_1: - Edges and flows: $u \rightarrow x_1$ (1,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (1,7), $x_1 \rightarrow x_3$ (0,1), $x_2 \rightarrow v$ (1,6), $x_3 \rightarrow v$ (1,2), $x_4 \rightarrow x_3$ (1,4). Flow F_2: - Edges and flows: $u \rightarrow x_1$ (3,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (3,7), $x_1 \rightarrow x_3$ (0,1), $x_2 \rightarrow v$ (3,6), $x_3 \rightarrow v$ (1,2), $x_4 \rightarrow x_3$ (1,4).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0102	146	—	—	—	
voloshin-hypergraph_ DIA_0103	151	—	—	—	
voloshin-hypergraph_ DIA_0104	155	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																										
voloshin-hypergraph_ DIA_0105	157	—	—	—	The diagram illustrates the relationship between a hypergraph $\mathcal{H} = (X, \mathcal{D})$, its incidence matrix $I(\mathcal{H})$, and its bipartite graph $B(\mathcal{H})$. Hypergraph $\mathcal{H}$: A set of 6 vertices $\{1, 2, 3, 4, 5, 6\}$ and 5 hyperedges D_1, D_2, D_3, D_4, D_5. Vertex 1 is in D_1. Vertices 1, 2, 3, 4, 5 are in D_2. Vertices 1, 2, 3, 5, 6 are in D_3. Vertices 1, 2, 3, 4 are in D_4. Vertices 1, 3, 4, 5, 6 are in D_5. Incidence Matrix $I(\mathcal{H})$: <table border="1"> <thead> <tr> <th></th><th>D_1</th><th>D_2</th><th>D_3</th><th>D_4</th><th>D_5</th></tr> </thead> <tbody> <tr> <th>1</th><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr> <tr> <th>2</th><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr> <tr> <th>3</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>4</th><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr> <tr> <th>5</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>6</th><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> </tbody> </table> Bipartite Graph $B(\mathcal{H})$: A bipartite graph with vertices $\{1, 2, 3, 4, 5, 6\}$ on the left and $\{D_1, D_2, D_3, D_4, D_5\}$ on the right. Edges connect vertices to the hyperedges they belong to: $(1, D_1), (1, D_2), (1, D_3), (2, D_2), (2, D_3), (2, D_4), (3, D_2), (3, D_3), (3, D_4), (3, D_5), (4, D_2), (4, D_3), (4, D_4), (4, D_5), (5, D_2), (5, D_3), (5, D_4), (5, D_5)$. Hypergraph $\mathcal{H}^*$: A set of 6 vertices $\{1, 2, 3, 4, 5, 6\}$ and 5 hyperedges d_1, d_2, d_3, d_4, d_5. Vertex 1 is in d_1. Vertices 1, 2, 3, 4, 5 are in d_2. Vertices 1, 2, 3, 5, 6 are in d_3. Vertices 1, 2, 3, 4 are in d_4. Vertices 1, 3, 4, 5, 6 are in d_5.		D_1	D_2	D_3	D_4	D_5	1	1	1	1	0	0	2	0	1	1	1	0	3	0	0	0	1	1	4	0	0	1	0	1	5	0	0	0	1	1	6	0	0	0	0	0
	D_1	D_2	D_3	D_4	D_5																																										
1	1	1	1	0	0																																										
2	0	1	1	1	0																																										
3	0	0	0	1	1																																										
4	0	0	1	0	1																																										
5	0	0	0	1	1																																										
6	0	0	0	0	0																																										
voloshin-hypergraph_ DIA_0106	158	—	—	—	The diagram illustrates the relationship between a graph G and its dual graph G^*. Graph G: A graph with 5 vertices $\{1, 2, 3, 4, 5\}$ and 5 edges a, b, c, d, e. Edges are $a = (1, 5), b = (1, 2), c = (2, 3), d = (2, 4), e = (5, 4)$. Dual Graph G^*: A graph with 5 vertices $\{a, b, c, d, e\}$ and 5 edges $1, 2, 3, 4, 5$. Edges are $1 = (a, b), 2 = (b, c), 3 = (c, d), 4 = (d, e), 5 = (e, a)$. The vertices a, b, c, d, e are arranged in a cycle, with c inside the cycle.																																										

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0107	161	—	—	—	K_4^0 K_4^1 K_4^2 K_4^3 K_4^4
voloshin-hypergraph_ DIA_0108	163	—	—	—	$\mathcal{H}$ $\mathcal{H}_1 = \mathcal{H} - x$
voloshin-hypergraph_ DIA_0109	163	—	—	—	$\mathcal{H}$ $\mathcal{H}_1 = \mathcal{H} - x$
voloshin-hypergraph_ DIA_0110	164	—	—	—	$\mathcal{H}$ $\mathcal{H}_1 = \mathcal{H} - D$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0112	165	—	—	—	
voloshin-hypergraph_ DIA_0113	166	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0115	171	—	—	—	
voloshin-hypergraph_ DIA_0116	172	—	—	—	

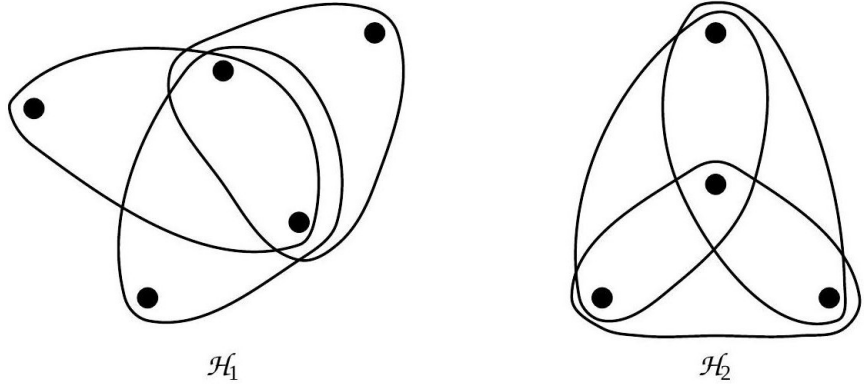
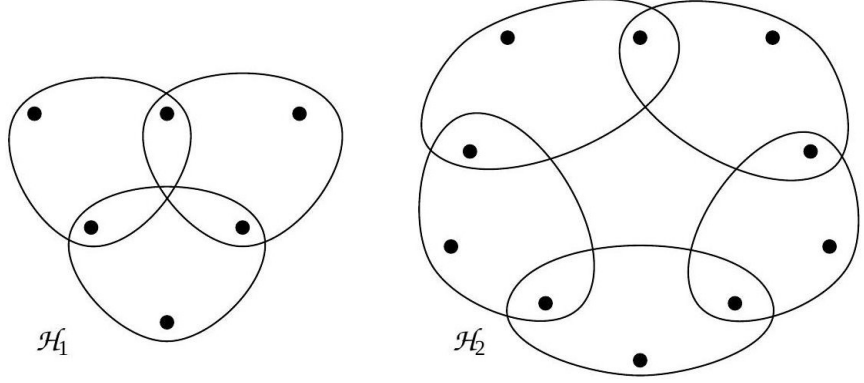
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voloshin-hypergraph_ DIA_0118	178	—	—	—	T C_4 T C_4

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0120	180	—	—	—	$\mathcal{H}$ $\mathcal{H}_0$
voloshin-hypergraph_ DIA_0121	181	—	—	—	$\mathcal{H}_0^*$ $\mathcal{H}^*$
voloshin-hypergraph_ DIA_0122	181	—	—	—	$\mathcal{H}_0^*$ $\mathcal{H}^*$
voloshin-hypergraph_ DIA_0123	182	—	—	—	G $\mathcal{H}$ $\mathcal{H}^*$

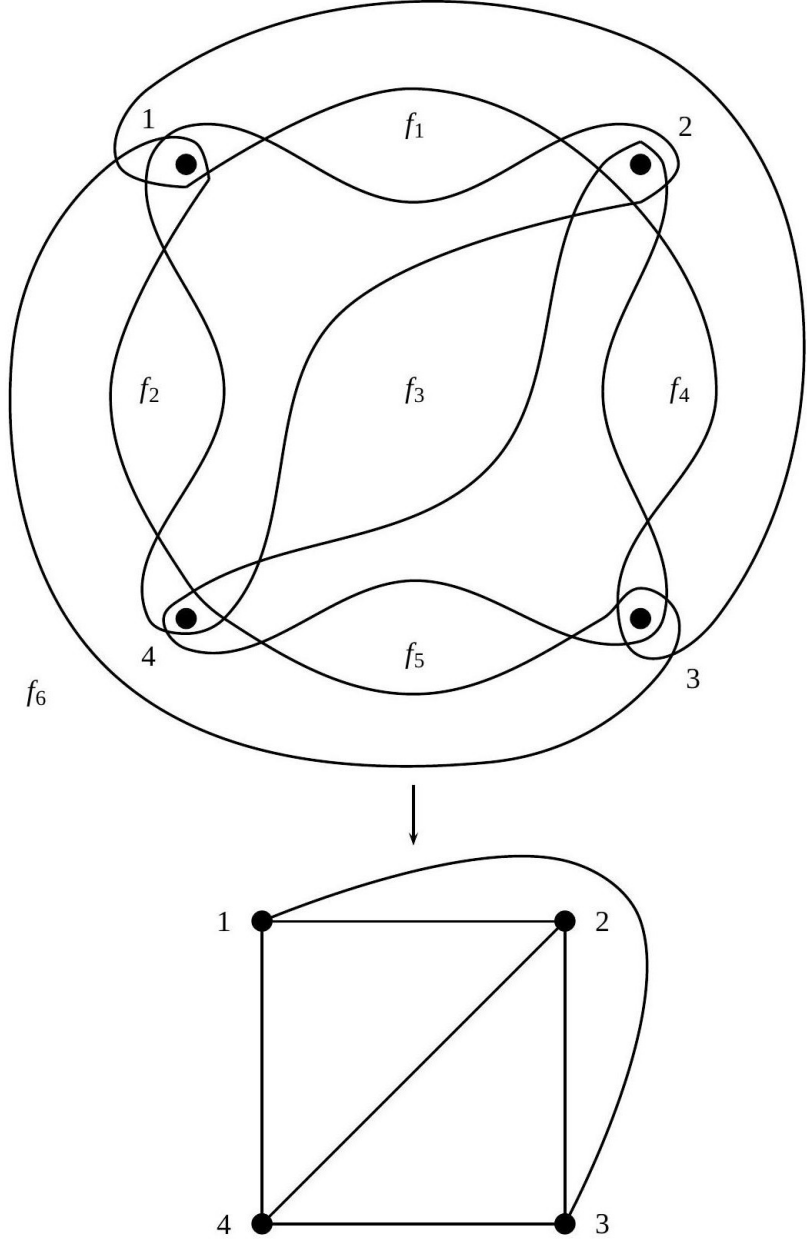
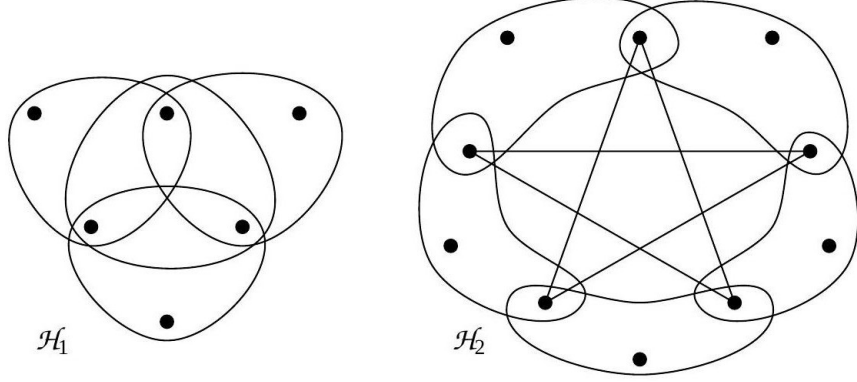
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voloshin-hypergraph_ DIA_0125	186	—	—	—	
voloshin-hypergraph_ DIA_0126	187	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0127	188	—	—	—	
voloshin-hypergraph_ DIA_0128	189	—	—	—	
voloshin-hypergraph_ DIA_0129	191	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0130	194	—	—	—	Diagram illustrating a hypergraph $\mathcal{H}$ and its relationship to a tree $T \subseteq [\mathcal{H}]_2$. The top part shows $\mathcal{H}$ with a vertex x and an arrow pointing to T. The bottom part shows $\mathcal{H} - x$ and an arrow pointing to $T - x$.
voloshin-hypergraph_ DIA_0131	196	—	—	—	Diagram illustrating two hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$.
voloshin-hypergraph_ DIA_0132	198	—	—	—	Diagram illustrating a hypergraph structure with multiple vertices and hyperedges.
voloshin-hypergraph_ DIA_0133	199	—	—	—	Diagram illustrating two hypergraph structures, one above the other.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0134	200	—	—	—	 Diagram showing two hypergraphs, $\mathcal{H}_1$ and $\mathcal{H}_2$. $\mathcal{H}_1$ has 5 vertices and 3 hyperedges. $\mathcal{H}_2$ has 4 vertices and 3 hyperedges.
voloshin-hypergraph_ DIA_0135	202	—	—	—	 Diagram showing two hypergraphs, $\mathcal{H}_1$ and $\mathcal{H}_2$. $\mathcal{H}_1$ has 6 vertices and 3 hyperedges. $\mathcal{H}_2$ has 10 vertices and 6 hyperedges.

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voloshin-hypergraph_ DIA_0136	203	—	—	—	

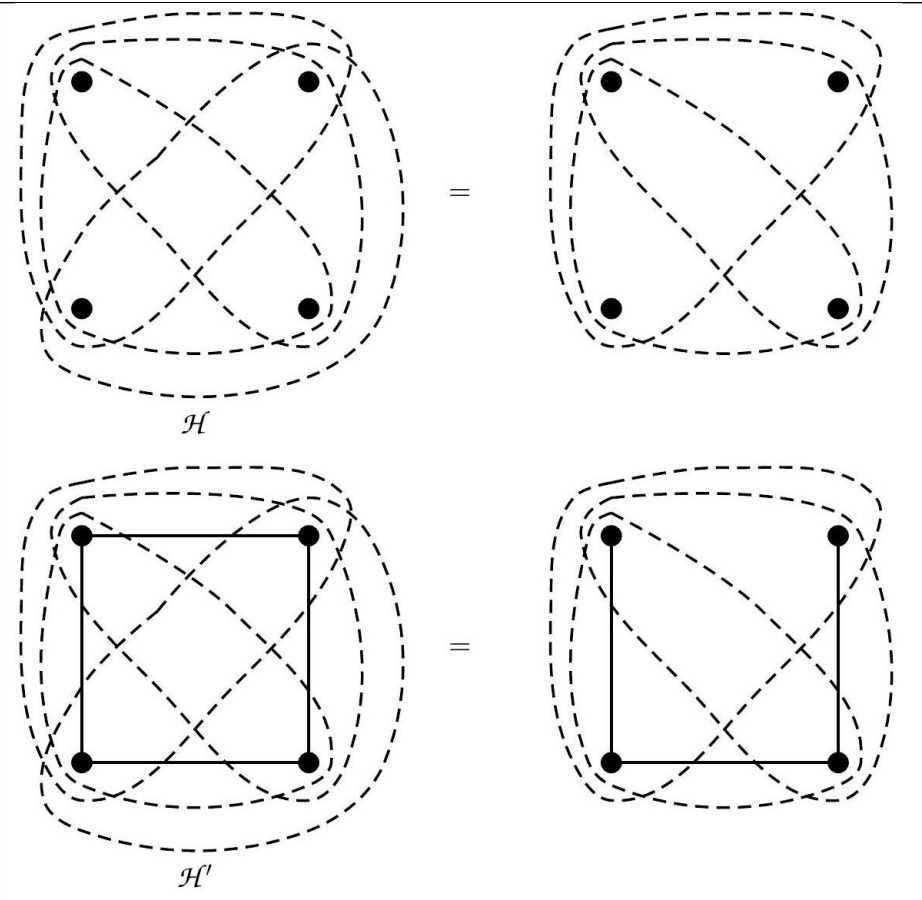
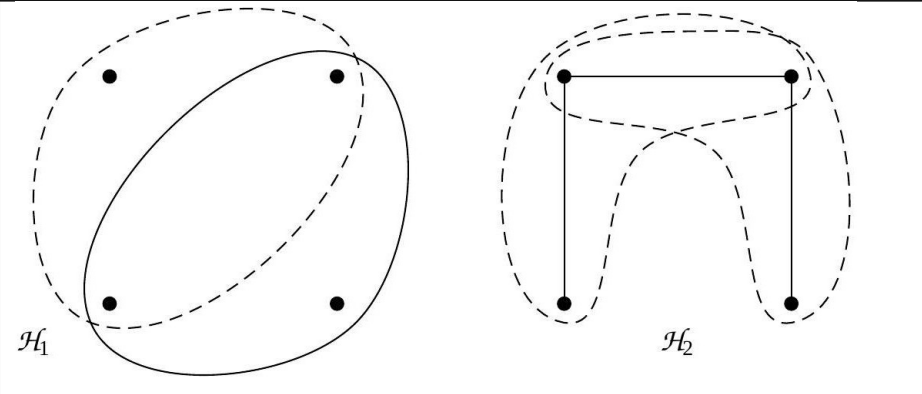
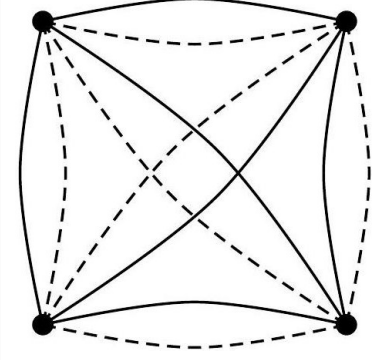
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0137	206	—	—	—	
voloshin-hypergraph_ DIA_0138	207	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0139	210	—	—	—	
voloshin-hypergraph_ DIA_0140	213	—	—	—	
voloshin-hypergraph_ DIA_0141	216	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0142	217	—	—	—	The diagram shows a large triangle with three vertices. Inside the triangle, there are three circles that overlap each other. There are four points: one at the top vertex, one at the bottom-left vertex, one at the bottom-right vertex, and one in the central region where all three circles overlap. The symbol $\mathcal{H}$ is placed to the left of the triangle.
voloshin-hypergraph_ DIA_0143	218	—	—	—	The diagram shows two points labeled 1 and 2. Between them are two paths: a dashed arc labeled C above and a solid arc labeled D below.
voloshin-hypergraph_ DIA_0144	221	—	—	—	The diagram shows five points labeled 1, 2, 3, 4, and 5. There are several overlapping regions: a dashed region C_1 containing points 5 and 4; a dashed region C_2 containing points 4 and 3; a solid region D_1 containing points 4 and 3; a solid region D_2 containing points 3 and 2; and a dashed region containing point 1. The symbol $\mathcal{H}$ is not explicitly shown.
voloshin-hypergraph_ DIA_0145	222	—	—	—	Two diagrams are shown side-by-side. The left diagram, labeled $\mathcal{H}_1$, shows a triangle with vertices and edges labeled D_1, D_2, and D_3, all enclosed within a dashed circle labeled C_1. The right diagram, labeled $\mathcal{H}_2$, shows a circle containing three points connected by dashed lines labeled C_1, C_2, and C_3, with the entire structure enclosed in a solid circle labeled D_1.
voloshin-hypergraph_ DIA_0146	223	—	—	—	The diagram shows four points arranged in a square pattern. There are two overlapping regions, each represented by a solid ellipse and a dashed ellipse. The symbol $\mathcal{H}$ is placed to the left of the diagram.

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voloshin-hypergraph_ DIA_0147	226	—	—	—	
voloshin-hypergraph_ DIA_0148	229	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0149	232	—	—	—	The diagram illustrates a sequence of hypergraph transformations: Top Row: A hypergraph C (a diamond shape with nodes 1, 2, 3, 4 and edges D_1, D_2, D_3, D_4) is shown. It is equated to the sum of two hypergraphs: one with a vertical edge D_1 and one with a vertical edge D_2. This step is labeled "splitting 13". Second Row: The two hypergraphs from the first row are each split further (labeled "spl. 24") into two more hypergraphs each. The sum of these four hypergraphs is equated to the sum of a triangle (1, 2, 3) with node 24 and a triangle (1, 2, 3) with node 13. This step is labeled "C-clearing + contraction 13". Third Row: The triangle (1, 2, 3) with node 24 is split (labeled "spl. 24") into two triangles: one with node 13 and one with node 24. The triangle (1, 2, 3) with node 13 is split (labeled "spl. 24") into two triangles: one with node 13 and one with node 24. This step is labeled "elimination". Bottom Row: The final result is the sum of three components: a triangle (1, 2, 3) with node 24, a triangle (1, 2, 3) with node 13, and a single node 13 with node 24. This step is labeled "C-clearing + contraction 24".

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voloshin-hypergraph_ DIA_0150	234	—	—	—	
voloshin-hypergraph_ DIA_0151	235	—	—	—	
voloshin-hypergraph_ DIA_0152	237	—	—	—	

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voloshin-hypergraph_ DIA_0153	238	—	—	—	
voloshin-hypergraph_ DIA_0154	239	—	—	—	
voloshin-hypergraph_ DIA_0155	242	—	—	—	
voloshin-hypergraph_ DIA_0156	242	—	—	—	$\mathcal{H}$

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voloshin-hypergraph_ DIA_0157	244	—	—	—	
voloshin-hypergraph_ DIA_0158	250	—	—	—	
voloshin-hypergraph_ DIA_0159	251	—	—	—	
voloshin-hypergraph_ DIA_0160	253	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0161	257	—	—	—	
voloshin-hypergraph_ DIA_0162	258	—	—	—	
voloshin-hypergraph_ DIA_0163	259	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0164	261	—	—	—	
voloshin-hypergraph_ DIA_0165	262	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0166	265	—	—	—	
voloshin-hypergraph_ DIA_0167	267	—	—	—	
voloshin-hypergraph_ DIA_0168	269	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0169	271	—	—	—	
voloshin-hypergraph_ DIA_0170	275	—	—	—	1st coloring: $ABBA$ 2nd coloring: $ABAB$ 3rd coloring: $AABB$ $\Rightarrow R(\mathcal{H}) = (0,3,0,0), P(\mathcal{H},\lambda) = 3\lambda(\lambda-1)$ $\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0171	276	—	—	—	
voloshin-hypergraph_ DIA_0172	277	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0173	280	—	—	—	The diagram illustrates a mapping G from a graph to a hypergraph. The top graph has nodes Z_1, Z_2, Z_3 and edges $\{a, c\}, \{a, b\}, \{b, c\}$. The bottom hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ has nodes a, b, c and hyperedges Z_1, Z_2, Z_3. A double-headed arrow labeled G connects the two structures.
voloshin-hypergraph_ DIA_0174	281	—	—	—	The diagram illustrates a mapping G from a bipartite graph to a hypergraph. The top graph has nodes x_1, x_2, x_3, x_4 (jobs) and y_1, y_2, y_3, y_4, y_5 (resources) with various edges. The bottom hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ has nodes x_1, x_2, x_3, x_4 and hyperedges Z_1, Z_2, Z_3. A double-headed arrow labeled G connects the two structures.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0175	284	—	—	—	Induction basis Induction hypothesis Induction step 1. P_1 2. $P_k \rightarrow P_{k+1}$ $\longrightarrow P_n$ is true for all $n = 1, 2, \dots$
voloshin-hypergraph_ DIA_0176	284	—	—	—	Induction basis Induction hypothesis Induction step 1. P_1 2. $P_1, P_2, \dots, P_k \rightarrow P_{k+1}$ $\longrightarrow P_n$ is true for all $n = 1, 2, \dots$